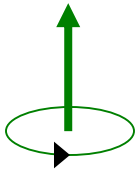
Magnetism and magnetic measurements

Magnetic moment of free currents

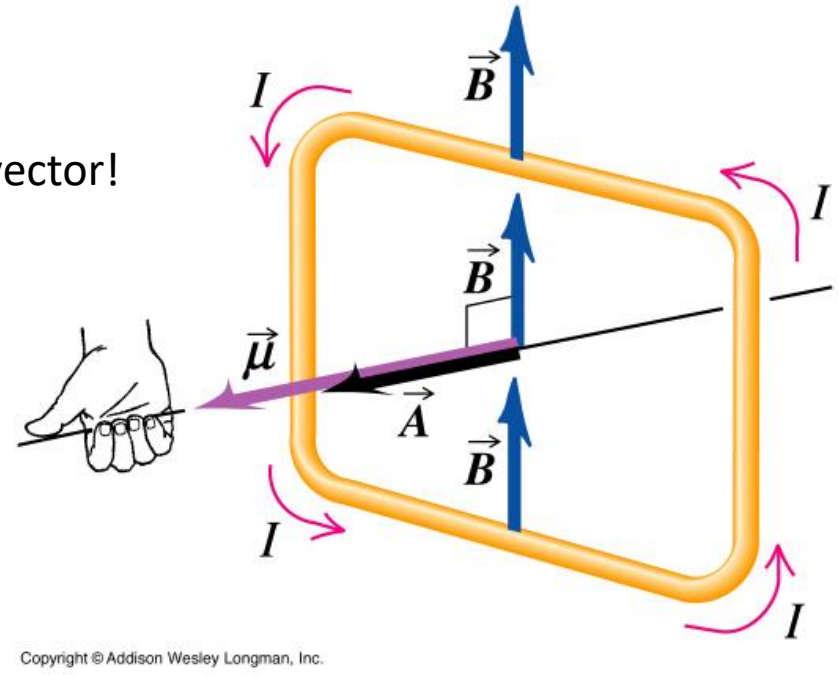
Magnetic moment of a closed loop carrying current I: $\mathbf{M}_i = \frac{I}{2c}\oint_C \mathbf{r}\times d\mathbf{l} = IS\mathbf{n}$



Magnetic field on the axis of a loop of radius R at a distance z is: $H_z = \frac{2M_i}{\left(R^2 + \ ^2\right)^{3/2}}$

Total magnetic moment: $\mathbf{M} = \sum \mathbf{M}_i$ (superposition principle)

Remember, M is a vector!



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Atomic and electronic moments

$$\mathbf{M}_{ion} = \gamma\hbar\mathbf{J} = -g\,\mu_B\mathbf{J}$$

$$\hbar\mathbf{J} = \hbar\mathbf{L} + \hbar\mathbf{S}$$

orbital spin

total angular momentum

γ - gyromagnetic ratio

g – Landé factor

$$\mu_B = \frac{e\hbar}{2mc} \approx 9.27410\times10^{-21}\ \frac{erg}{G}$$ — Bohr magneton

$$g = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+\ )}$$

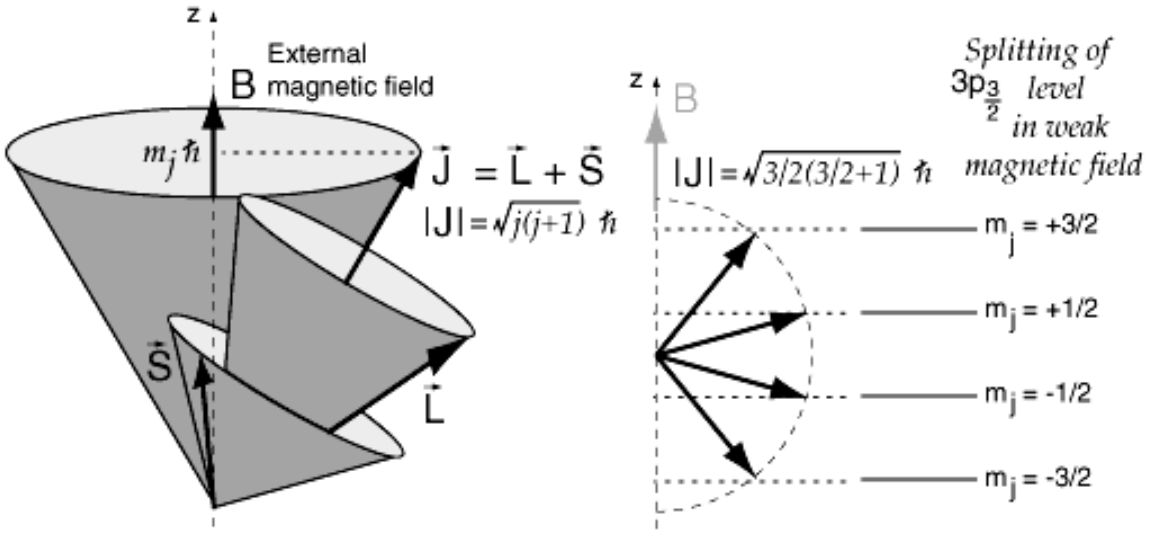
free electron:

$$g = 2.0023 \approx 2.00$$

Magnetic moment:

$$M_e \approx \mu$$

(J=S=1/2)



Magnetic moment in numbers

it measures a total magnetic moment in cgs (emu)

1 emu is:

- M of a 1 m^2 loop carrying a 1 mA current
- M of a loop of radius 1.78 cm carrying a 1 A current
- Typical permanent magnet (1 mm^3) ~ 1 emu

- M of a neutron star ~ 10^{30} emu
- The Earth's magnetic moment ~ $8x10^{25}$ emu
- An electron spin: μ_B~10^{-20} emu
- Proton and neutron: μ_N~10^{-23} emu

- One Abrikosov vortex (0.1 mm long) ~ 10^{-10} emu
- Change in M due to d-wave gap < 10^{-10} emu/K
- Hard superconductors ~ 0.1 emu

Basics

$$B\left[\text{Gauss}\right] = H\left[\text{Oersted}\right] + 4\pi M\left[\frac{\text{emu}}{\text{cm}^3}\right],$$

total magnetic moment of a sample of volume V

$$B = \mu$$

"magnetization" → $$M = \frac{m}{V}\left[\frac{\text{erg}}{\text{G}\cdot\text{cm}^3}\right]$$

$$\left.\begin{aligned} B &= \mu H \\ M &= \chi H \end{aligned}\right\} \Rightarrow \mu H = H + 4\pi\chi H \Rightarrow$$

$$\Rightarrow \mu = 1 + 4\pi\chi$$

magnetic permeability

magnetic susceptibility

this is ONLY true for homogeneous, uniform para- or dia- magnetic systems

In SI

$$B\left[\text{Tesla}\right] = H\left[\frac{A}{m}\right] + m\left[A \cdot m\right],$$

$$B = \mu$$

$$m = M\left\lfloor \frac{\text{erg}}{\text{G}} \right\rfloor / V$$

$$\left.\begin{aligned} B &= \mu H \\ m &= \chi H \end{aligned}\right\} \Rightarrow \mu H = H + \chi H \Rightarrow$$

$$\Rightarrow \mu = 1 + \chi$$

magnetic permeability

magnetic susceptibility

Magnetic Susceptibility

Magnetic moment

$$1\left[\mathrm{Amp}\cdot\mathrm{m}^2\right] = 1\left[\mathrm{J}\right] = 10^{\left[\frac{\mathrm{erg}}{\mathrm{G}}\right]}$$

Magnetic field

$$1\left[\frac{\mathrm{A}}{m}\right] = \frac{4\pi}{10^3}\left[Oe\right] \approx 0.013\left[Oe\right], \quad 1\left[Oe\right] = \frac{10^3}{4\pi}\left[\frac{\mathrm{A}}{m}\right] \approx 80\left[\frac{\mathrm{A}}{m}\right]$$

Magnetic induction

$$1\left[T\right] = 10^4\left[G\right]$$

NOTE: QD MPMS measures M [emu], not 4πM

Magnetic Susceptibility

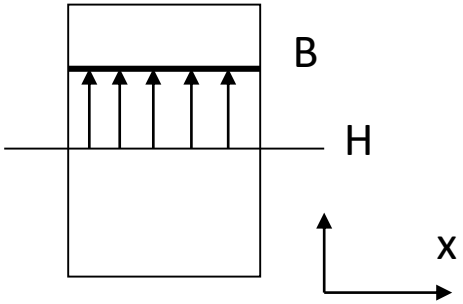
$\chi = m / H$ - dimensionless! $\chi_{SI} = 4\pi\chi_{cgs}$

$$m = M \left\lfloor \frac{\text{erg}}{\text{G}} \right\rfloor / V$$

some other quantities are used:

χ_m $\quad \frac{\chi}{\rho} \left\lfloor \frac{cc}{cc \cdot g} = g^{-1} \right|$

χ_{mol} $\quad \chi_m M_m \left\lfloor \frac{g}{g \cdot mol} = mol^{-1} \right|$



non-magnetic: $\chi = 0,\ \mu = 1$

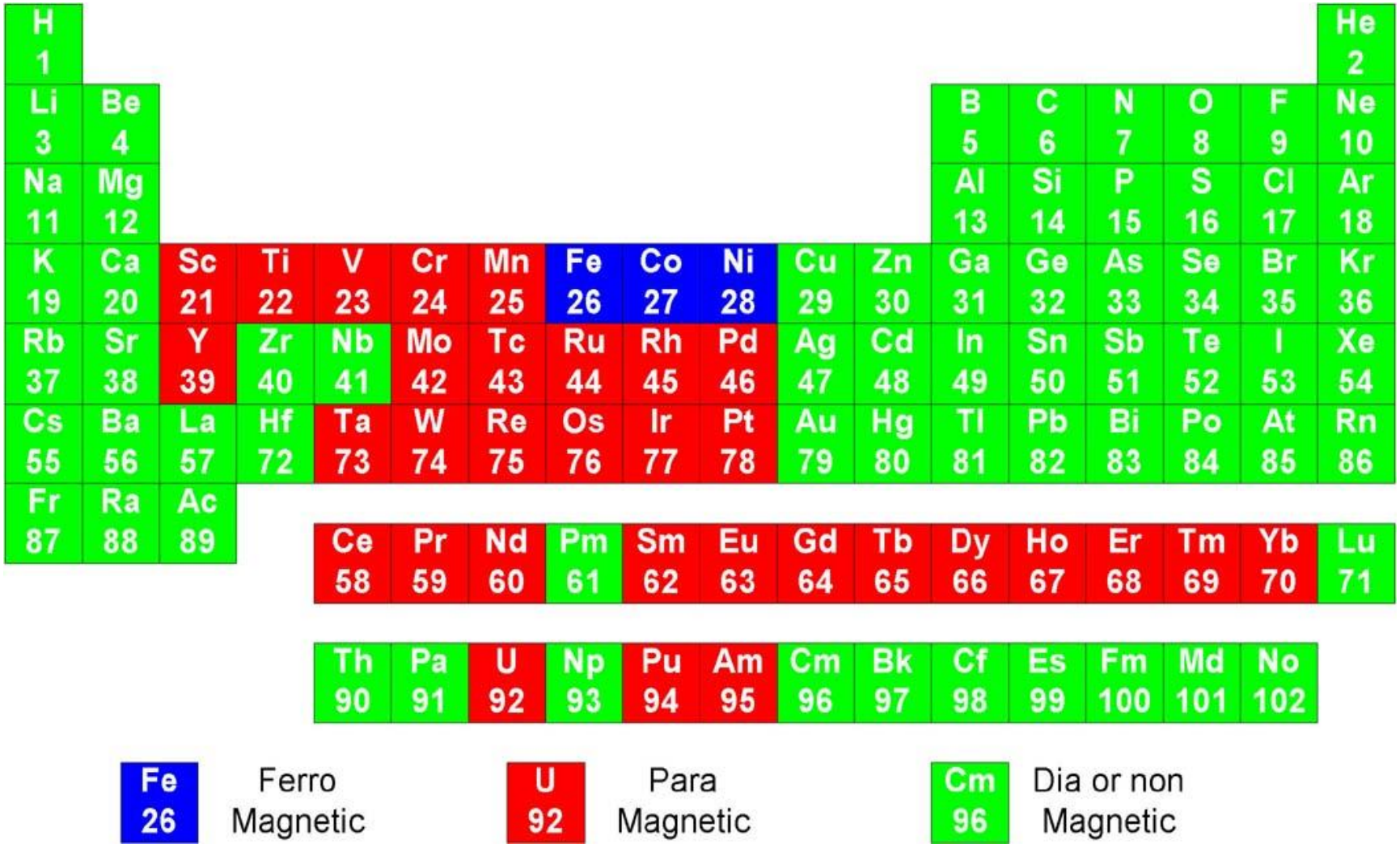
*para*magnetic: $\chi > 0,\ \mu > 1$

*dia*magnetic: $\chi < 0,\ \mu < 1$

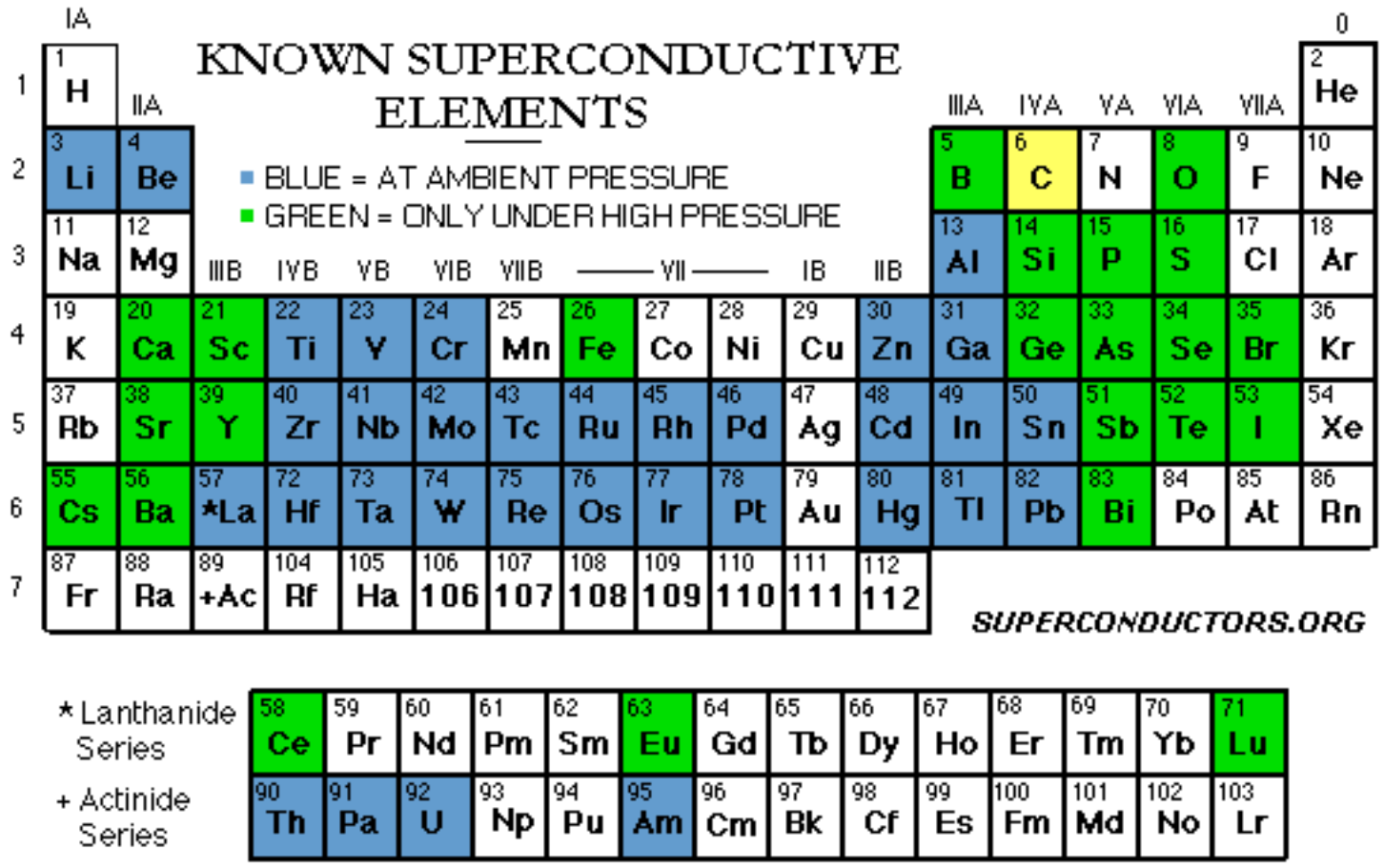
$\mu = 4\pi\chi +$

perfect diamagnetism: $\chi = -\frac{1}{4\pi},\ \mu = 0$

Magnetism in a Periodic Table



Superconductivity in a Periodic Table



General Form of Magnetic Susceptibility

$$H = H_{dc} + H_{ac}\cos(\omega t)$$

$$M(t) = M_0 + H_{ac}\sum_{n=1}^{\infty}\left[\chi_n'(\omega)\cos(n\omega t) + \chi_n''(\omega)\sin(n\omega t)\right]$$

where

$$\chi_n'(\omega) = \frac{\omega}{\pi H_{ac}}\int_{-\pi/\omega}^{\pi/\omega} M(t)\cos(n\omega t)\,dt$$

$$\chi_n''(\omega) = \frac{\omega}{\pi H_{ac}}\int_{-\pi/\omega}^{\pi/\omega} M(t)\sin(n\omega t)\,dt$$

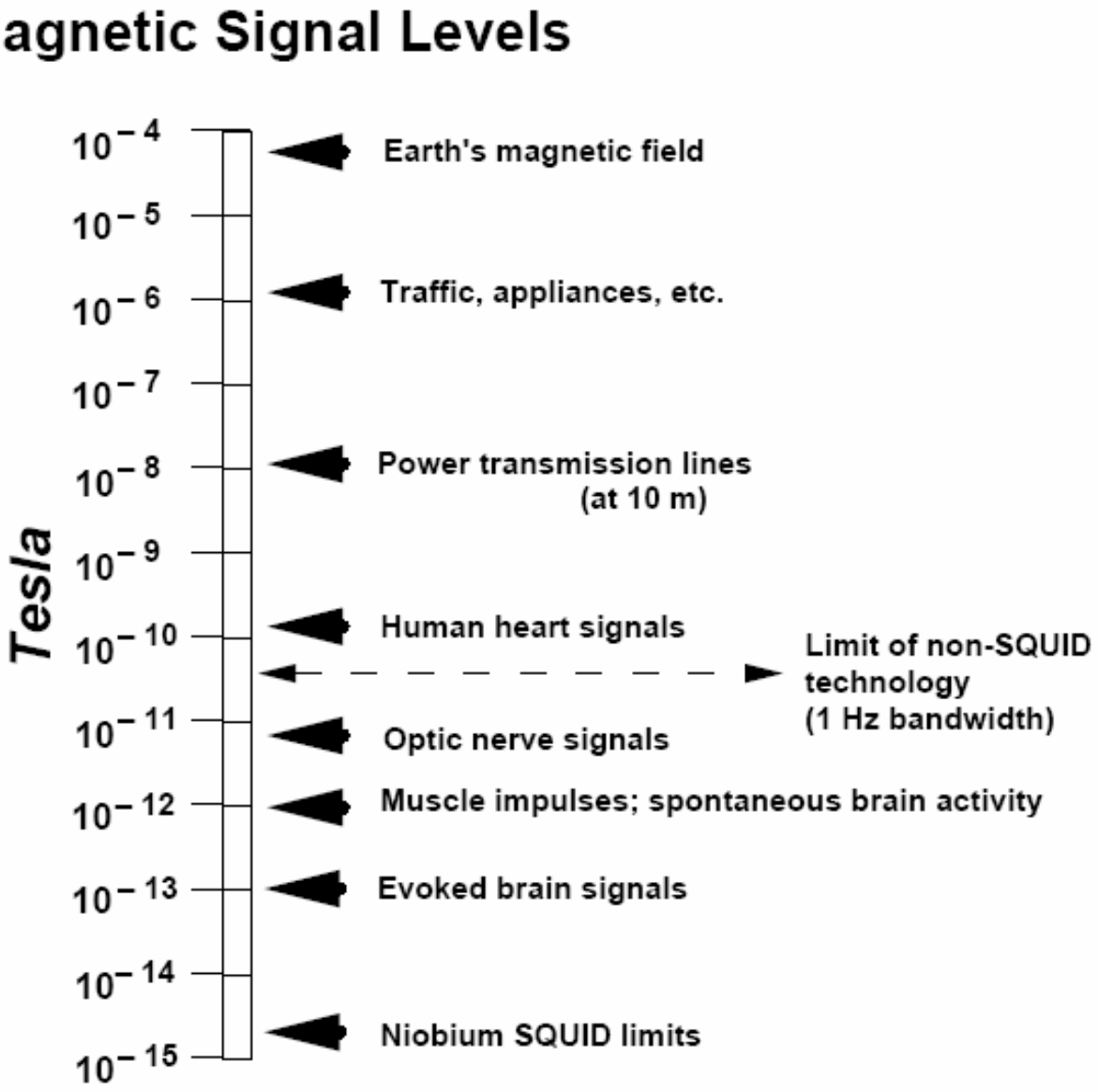
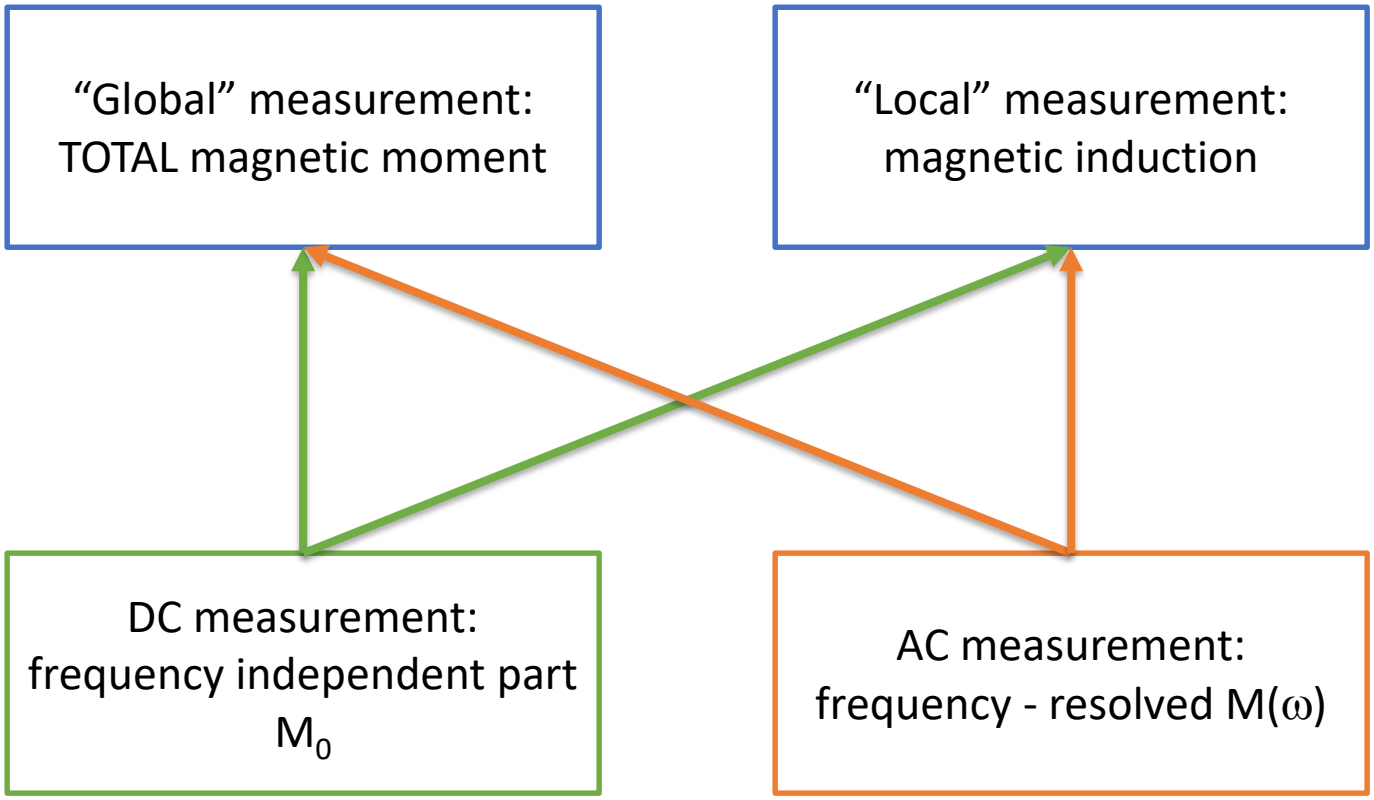


Figure 5-10 Magnetic field levels of various sources.


“Global” measurement:
TOTAL magnetic moment
“Local” measurement:
magnetic induction
DC measurement:
frequency independent part
M_0
AC measurement:
frequency - resolved M(ω)

What is measured magnetic moment?

more specifically, - does it depend on the sample shape?

How to evaluate intrinsic magnetic susceptibility, χ, from the measurements of the total magnetic moment, m?

Definitions (all SI)

In local and magnetically linear media without demagnetization effects infinite slab or cylinder in parallel magnetic field),

$$B = \mu H = \mu_0 (m + H) \tag{1}$$

$$m = \frac{B}{\mu_0} - H = \chi H \tag{2}$$

$\rightarrow \mu_0 = 4\pi \times 10^{-7}$ [N/A^2 or H/m] is magnetic permeability of free space
$\rightarrow \mu$ and χ are total linear magnetic permeability and susceptibility

(In general, these are 2nd rank tensors. We consider isotropic case).

It follows then that,

$$\chi = \frac{\mu}{\mu_0} - 1 = \mu_r - 1 \tag{3}$$

where $\mu_r = \mu/\mu_0$ is relative magnetic permeability
($\mu_r = 1$ for non-magnetic media and $\mu_r = 0$ for a perfect diamagnet).

It follows then that,

$$\chi = \frac{\mu}{\mu_0} - 1 = \mu_r - 1 \tag{1}$$

where $\mu_r = \mu/\mu_0$ is relative magnetic permeability ($\mu_r = 1$ for non-magnetic media and $\mu_r = 0$ for a perfect diamagnet).

For finite samples of ellipsoidal shape, *constant* demagnetization factors, N, connect the applied magnetic field H_0 with the internal field, H,

$$H = H_0 - Nm \tag{2}$$

and in terms of an applied field the magnetization is:

$$m = \frac{\chi}{1 + \chi N} H_0 \tag{3}$$

Effective Demagnetization Factors

We now *define* effective magnetic susceptibility, χ_0 with "intrinsic" (no demag) susceptibility, χ, with *applied* magnetic field, H_0,

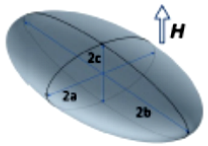
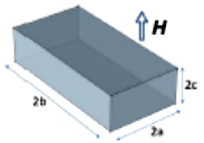
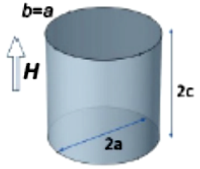
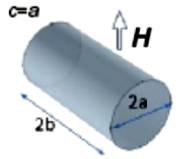
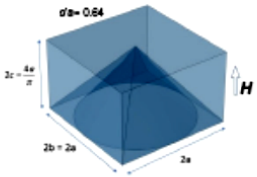
$$M = \chi_0 H_0 V = \frac{\chi H_0 V}{1 + \chi N} \tag{1}$$

from which:

$$N = \frac{1}{\chi_0} - \frac{1}{\chi} \tag{2}$$

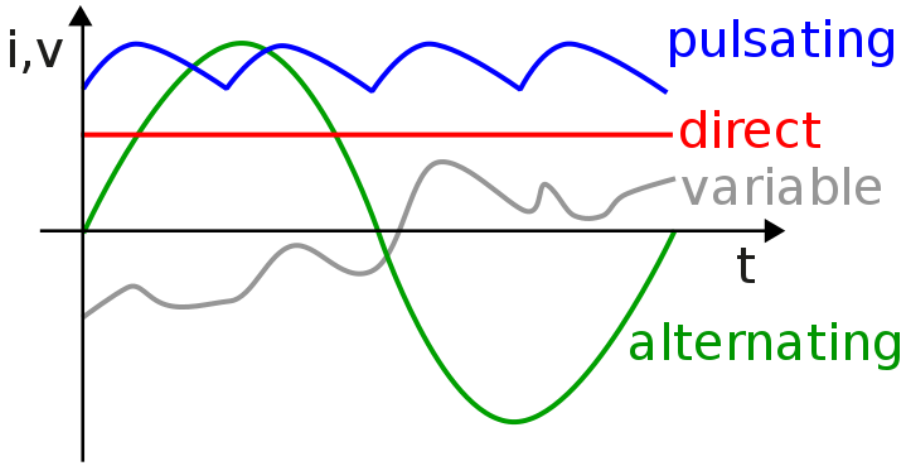
where $\chi \geq -1$ and $-\infty \leq \chi_0 \leq \infty$

TABLE I. The approximate effective ("integral") demagnetizing factor along an applied magnetic field.

Shape	Geometry	Demagnetizing factor along applied field
Ellipsoid (exact)		$N = \frac{1}{2}\frac{b}{a}\frac{c}{a}\int_0^\infty \frac{ds}{\left(s+\frac{c^2}{a^2}\right)\sqrt{(s+1)\left(s+\frac{b^2}{a^2}\right)\left(s+\frac{c^2}{a^2}\right)}}$
Rectangular cuboid		$N^{-1} = 1 + \frac{3}{4}\frac{c}{a}(1 + \frac{a}{b})$
Strip, rectangular		$N^{-1} = 1 + \frac{2}{3}\frac{c}{a}$ for $(c/a \gtrsim 5), = 1 + \frac{3}{4}\frac{c}{a}$ for $(c/a \lesssim 5)$
Strip, elliptical		$N^{-1} = 1 + \frac{c}{a}$
Cylinder, axial		$N^{-1} = 1 + 1.6\frac{c}{a}$
Cylinder, transverse		$N^{-1} = 2 + \frac{1}{\sqrt{2}}\frac{a}{b}$
Great pyramid (cone, cuboid)		$N = 0.64\ (0.64, 0.49)$

“DC” = Direct Current

“AC” = Alternating Current

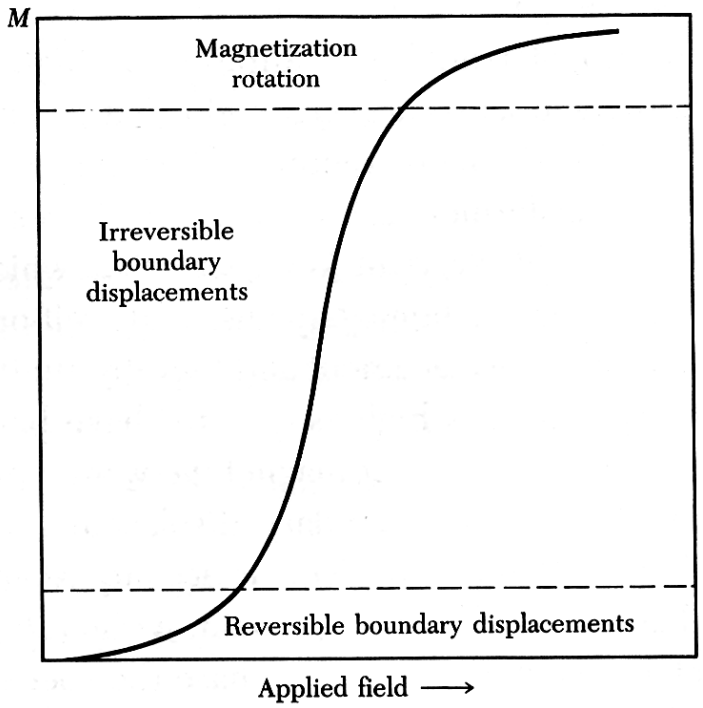


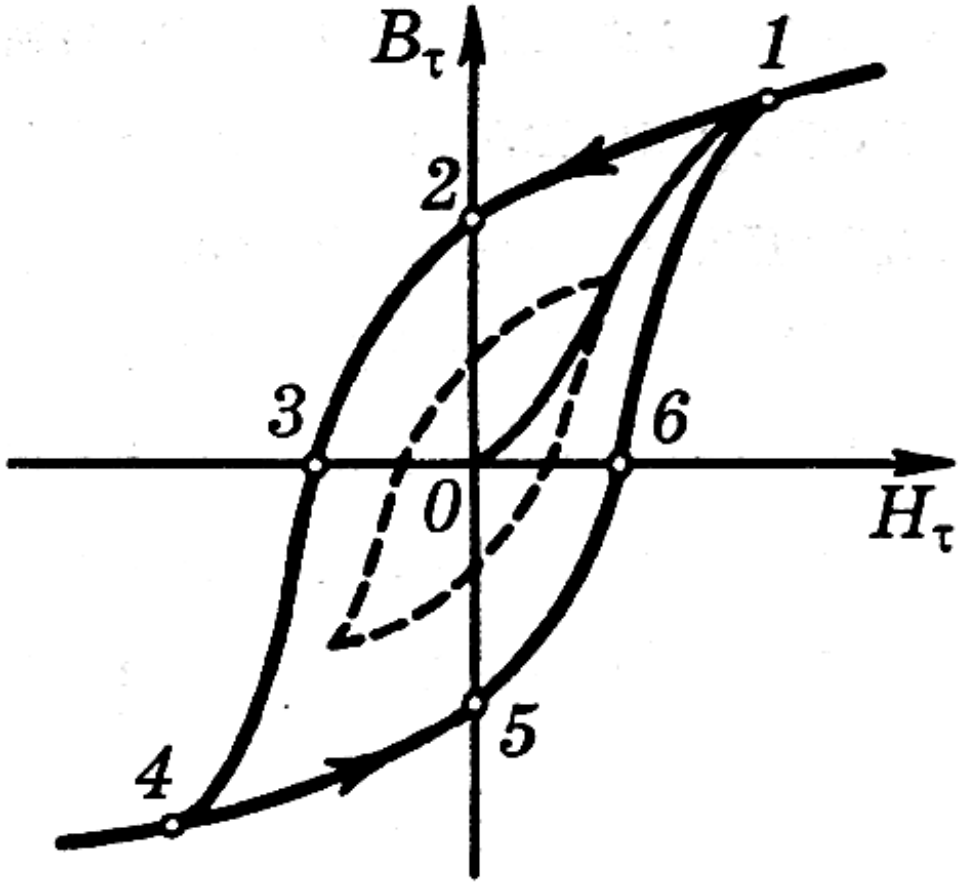
https://en.wikipedia.org/wiki/Direct_current#/media/File:Types_of_current.svg

“DC” Magnetization

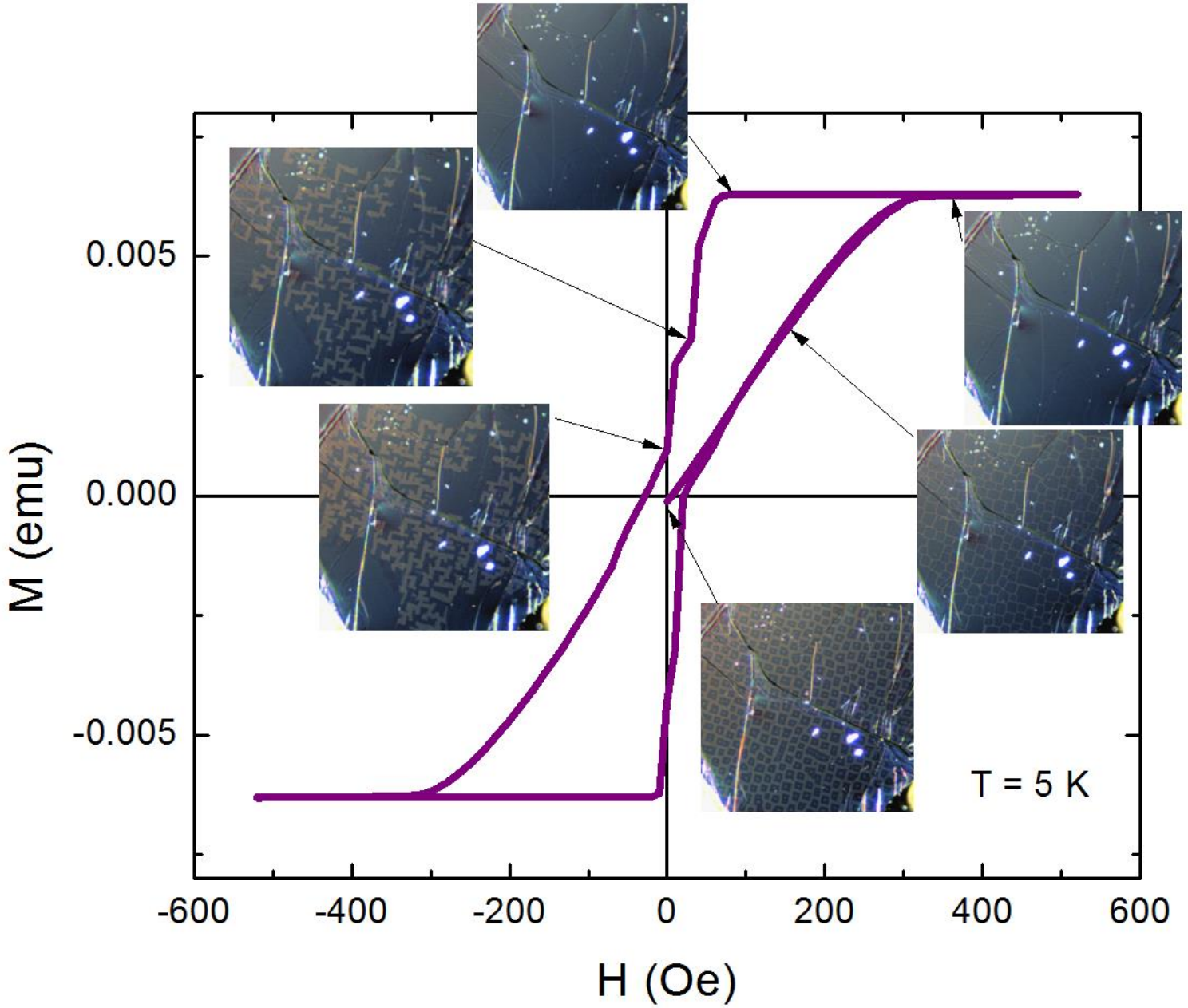
magnetic moment does not change (much) during the time window of the measurement

“Hysteresis” M(H) loops - Ferromagnet

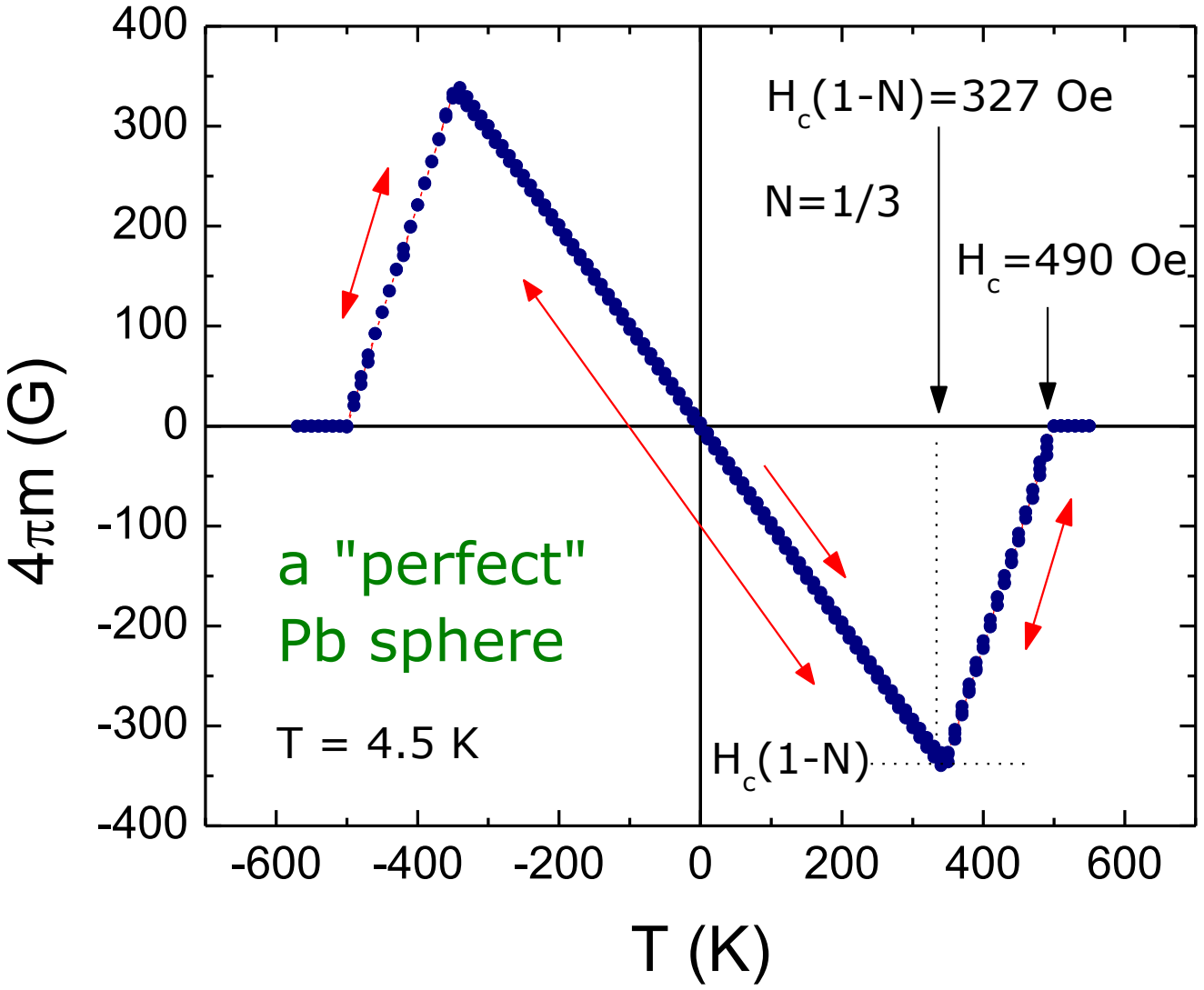


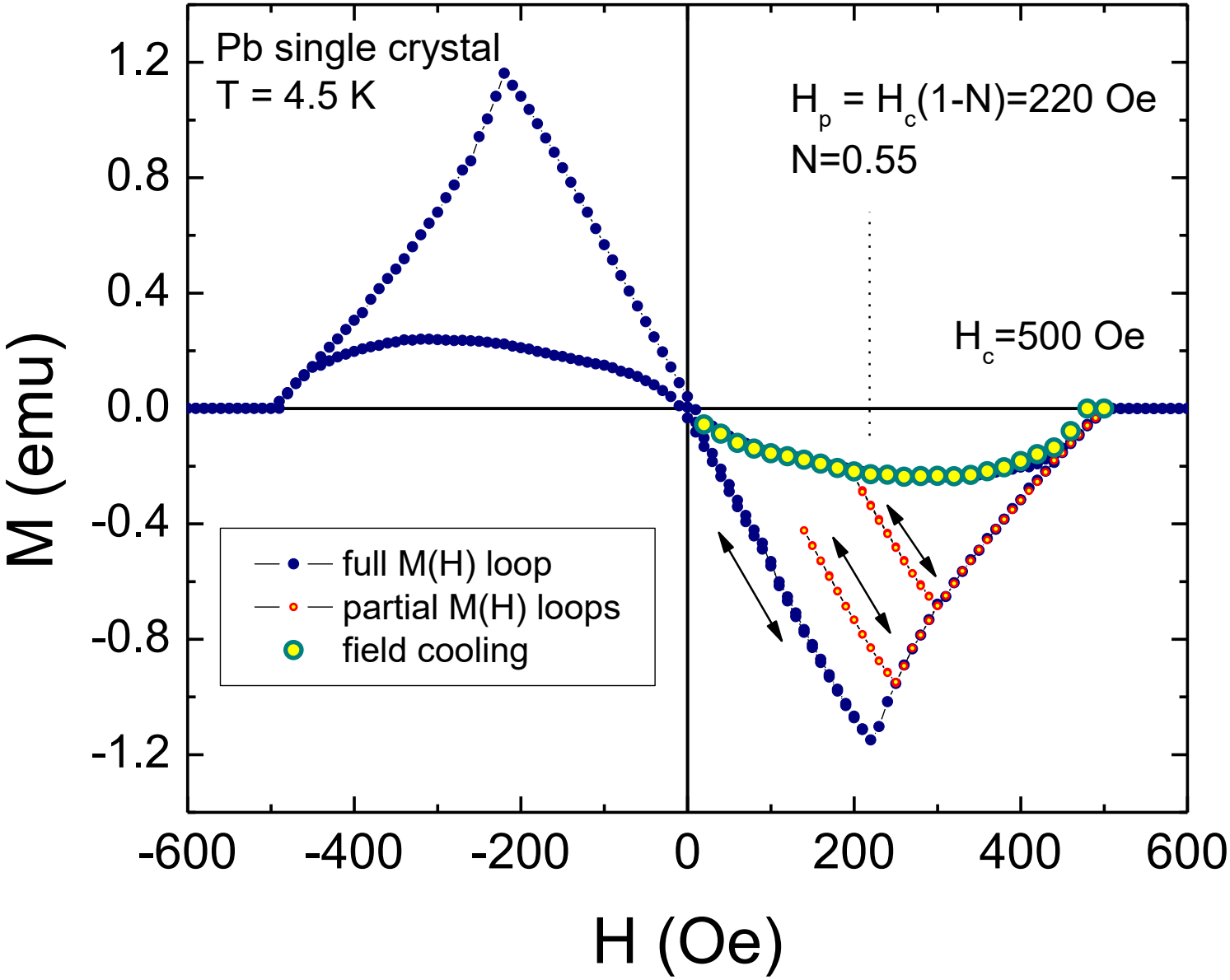


Influence of Domain Structure

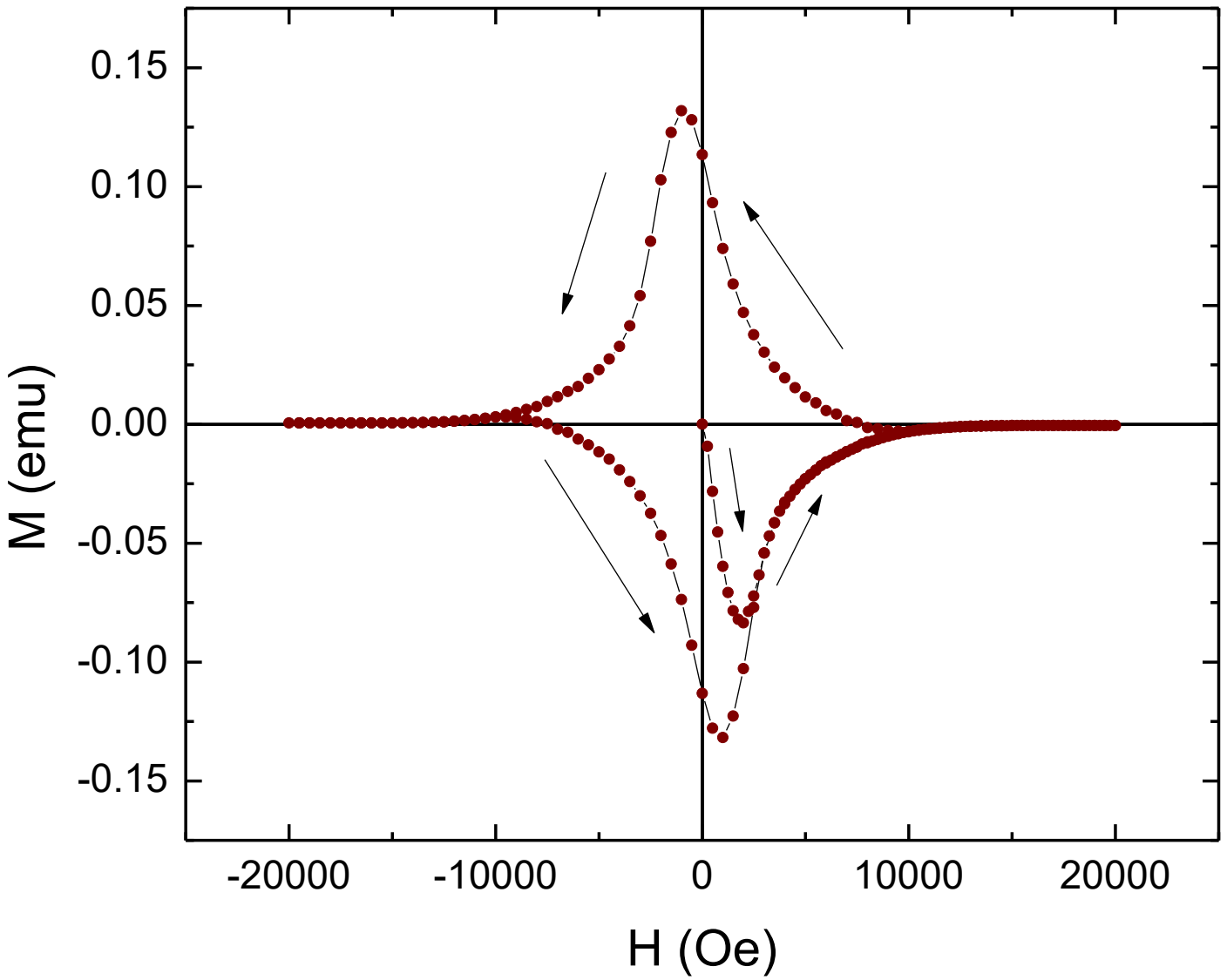


Type-I Superconductor




Pb single crystal
T = 4.5 K
H_p = H_c(1-N)=220 Oe
N=0.55
H_c=500 Oe
full M(H) loop
partial M(H) loops
field cooling
M (emu)
H (Oe)
1.2
0.8
0.4
0.0
-0.4
-0.8
-1.2
-600
-400
-200
0
200
400
600

typical type-II superconductor



Magnetometer

Popular definition (Wikipedia):

A **magnetometer** is a scientific instrument used to measure the strength and/or direction of the magnetic field in the vicinity of the instrument. Magnetism varies from place to place and differences in Earth's magnetic field (the magnetosphere) can be caused by the differing nature of rocks and the interaction between charged particles from the Sun and the magnetosphere of a planet. Magnetometers are often a frequent component instrument on spacecraft that explore planets.

Rotating coil magnetometer
Hall effect magnetometer
Proton precession magnetometer
Gradiometer
Fluxgate magnetometer

Lab definition:

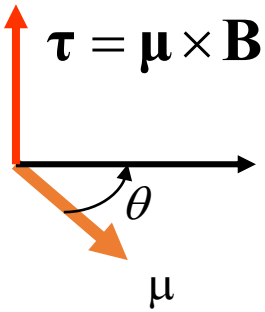
A device to measure magnetic moment or magnetic induction of finite – volume samples at fixed temperature and magnetic field. Magnetic moment is a vector, but only one component is measured at a time.

Types of GLOBAL Magnetic Measurements

- Use Faraday law of induction, $V = -\frac{d\Phi}{dt}$
  - extraction from the coil
  - MPMS SQUID
  - Vibrating sample magnetometer (VSM)
  - AC susceptibility
  - pulse

- Use the force or torque
  - Faraday balance (force)
  - torque magnetometer

torque:



B

Energy: $W = -\boldsymbol{\mu}\mathbf{B} = -\mu B\cos(\theta)$

Force: $F = -grad(W) = grad(\boldsymbol{\mu}\mathbf{B})$

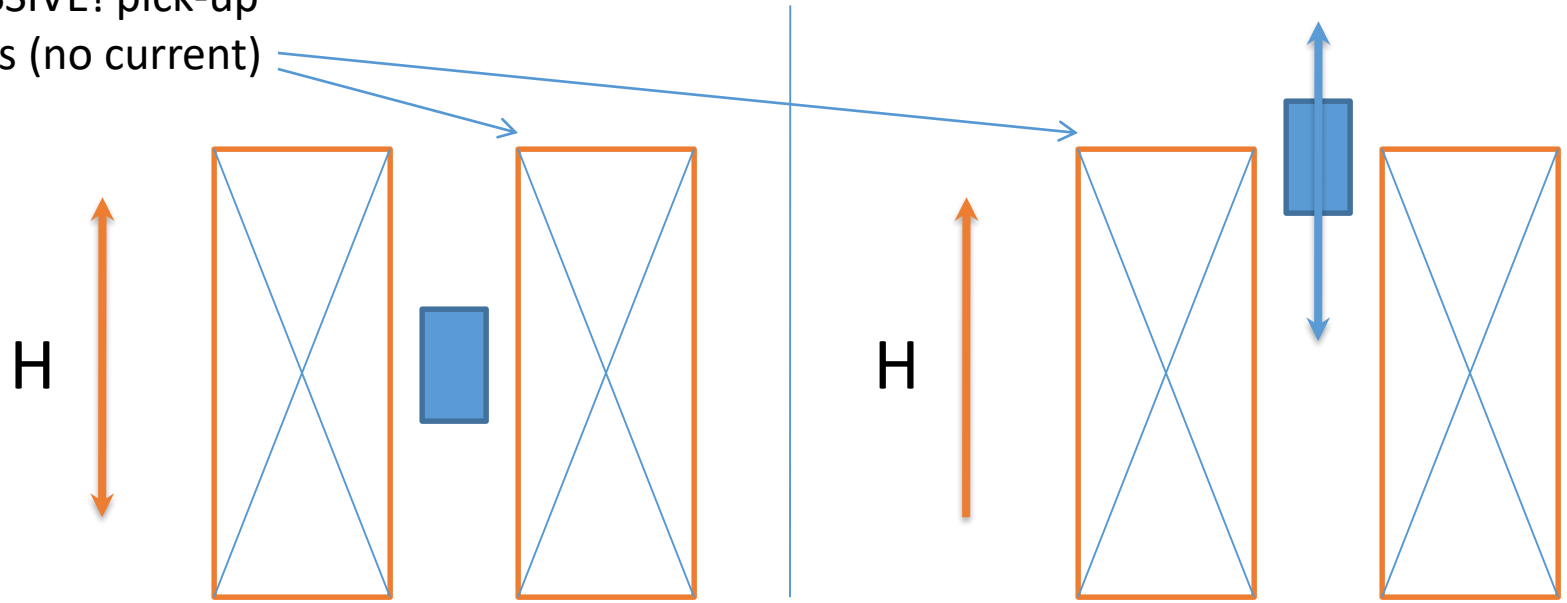
- Use sample as a nonlinear element of the cirquit
  - cavity perturbation (MW)
  - tunnel – diode resonator (RF)

Local Probes

- Surface probes
  - Kerr – effect magneto-optics
  - Faraday – effect magneto-optics
  - miniature Hall-probes
  - Bitter decoration
  - magnetic force microscope
  - miniature SQUID
- Bulk local probes
  - muon spin rotation
  - polarized neutron reflectometry
  - nuclear magnetic resonance
  - neutron scattering
- Menascale probes
  - mechanical resonant imaging
  - nanoSQUIDs
  - fluorescent confocal microscope
  - optical magnetometry of NV-centers in diamonds

AC/DC measurements

PASSIVE! pick-up coils (no current)



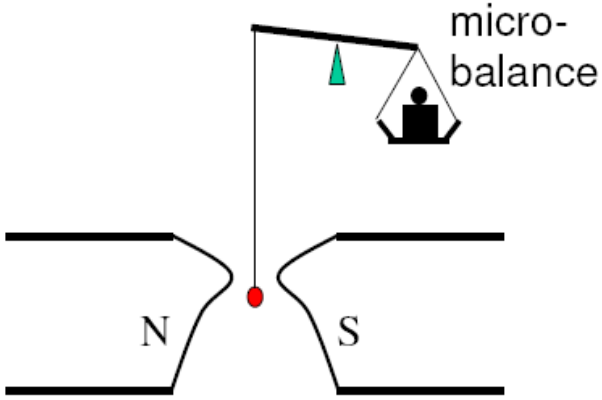
M(t)

M=const

$$V_{ac} = -\frac{\partial \Phi}{\partial t}$$

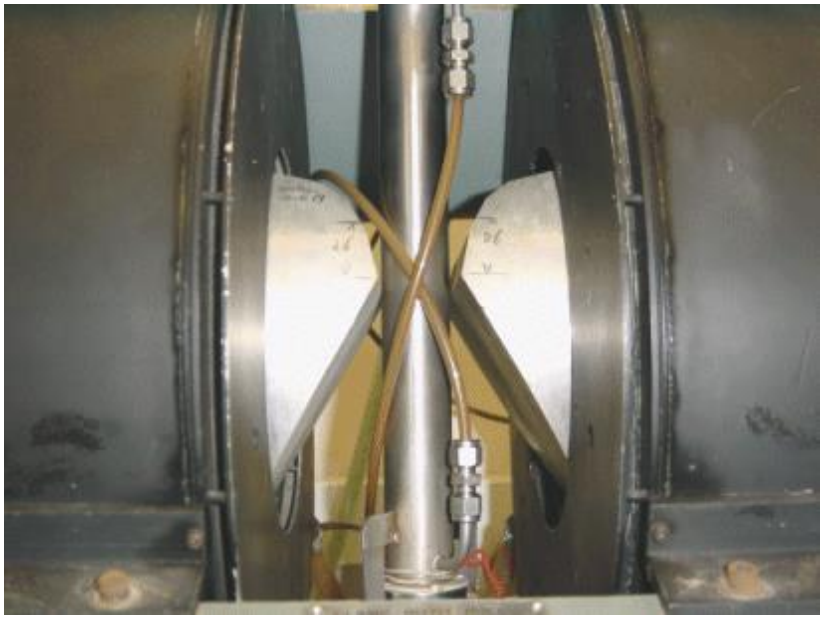
Lock-In Amplifier

Use the force – Faraday balance magnetometer



Faraday balance

Faraday pole caps have the property that in vertical direction z on the symmetry axis of the magnet, where the field, let us say, is in x-direction, the product $B_x * dB_x/dz$ is constant over a considerable range in z.



$$U = -\mathbf{MB}$$

$$\mathbf{F} = \nabla(\mathbf{MB})$$

if $\mathbf{B} = \left(B_x(\ ), 0, 0\right)$

$$F_z = \frac{d}{dz}\left(M_x B_x\right) = \chi \frac{d}{dz}\left(B_x^2\right) = \chi\left(2B_x \frac{dB_x}{dz}\right)$$

examples

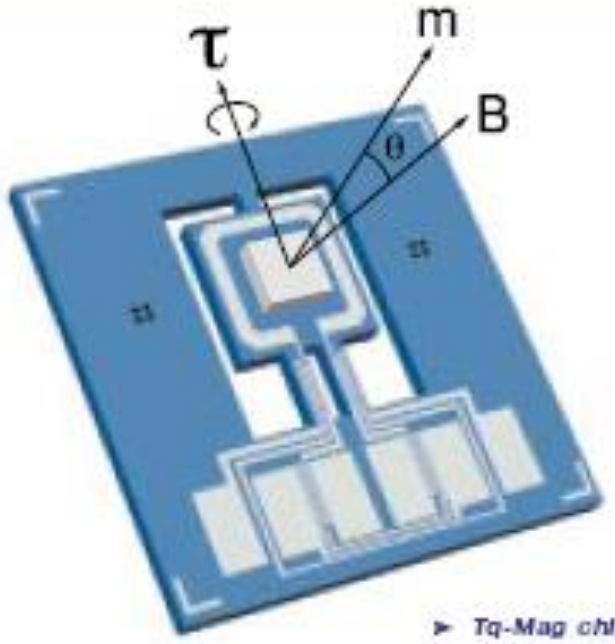


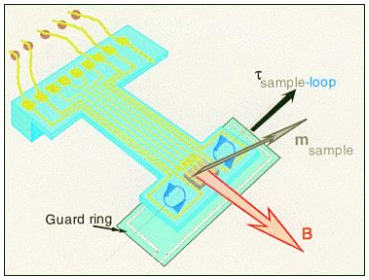






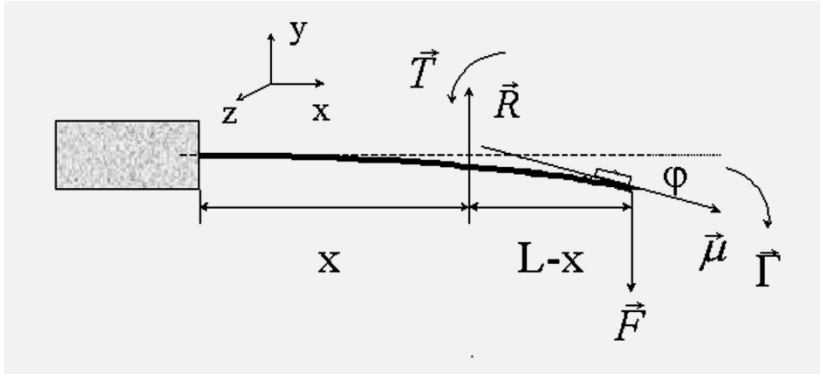
torque magnetometer





advantages?
- fast
- small samples

$$\boldsymbol{\tau} = \mathbf{M} \times \mathbf{B}$$



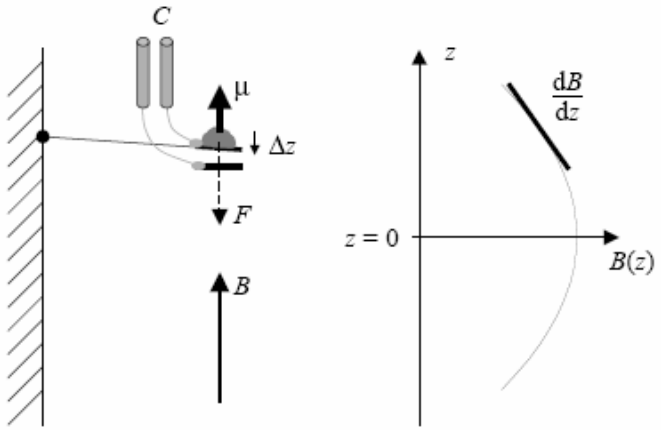
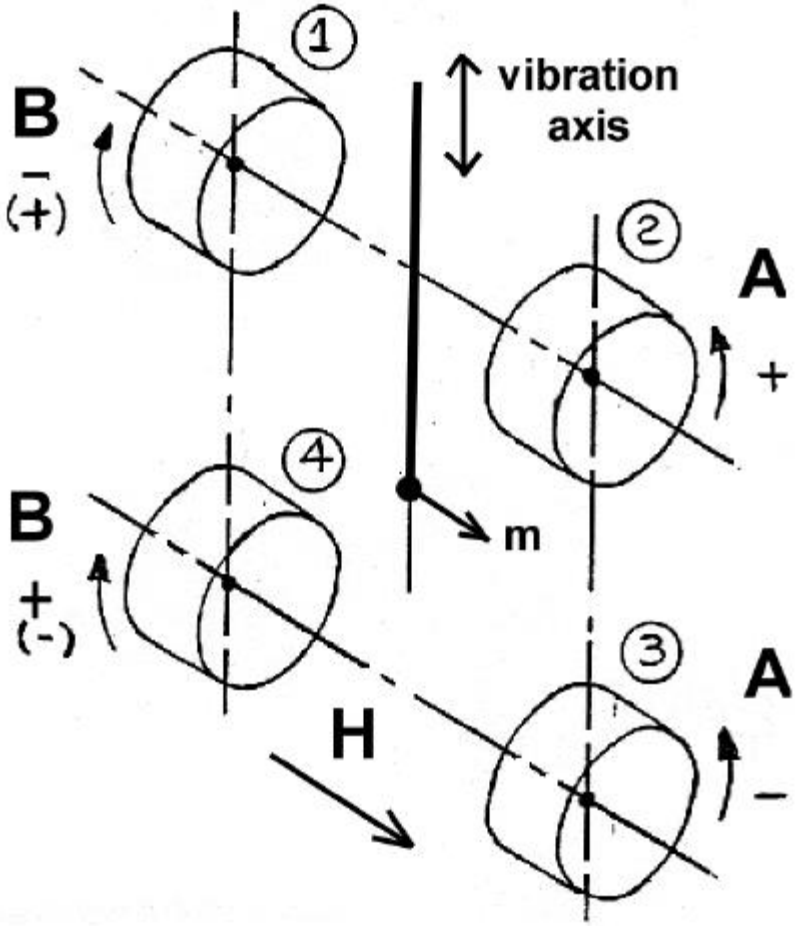


Fig. 2.14 : Scheme of a typical configuration for capacitive detection of the magnetic force on a paramagnetic isotropic sample in a field gradient. The magnitude of the field inhomogeneity is largely exaggerated.

Vibrating-sample magnetometer



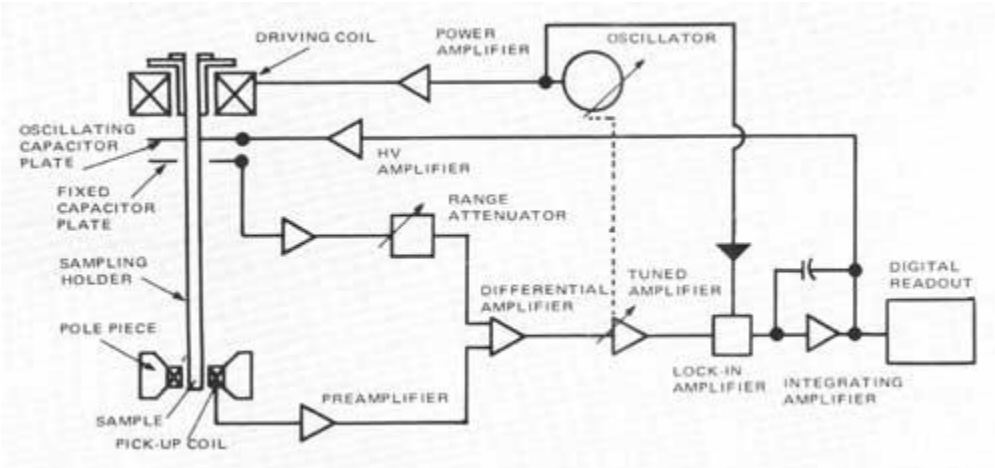


Figure 63. VSM detection coils

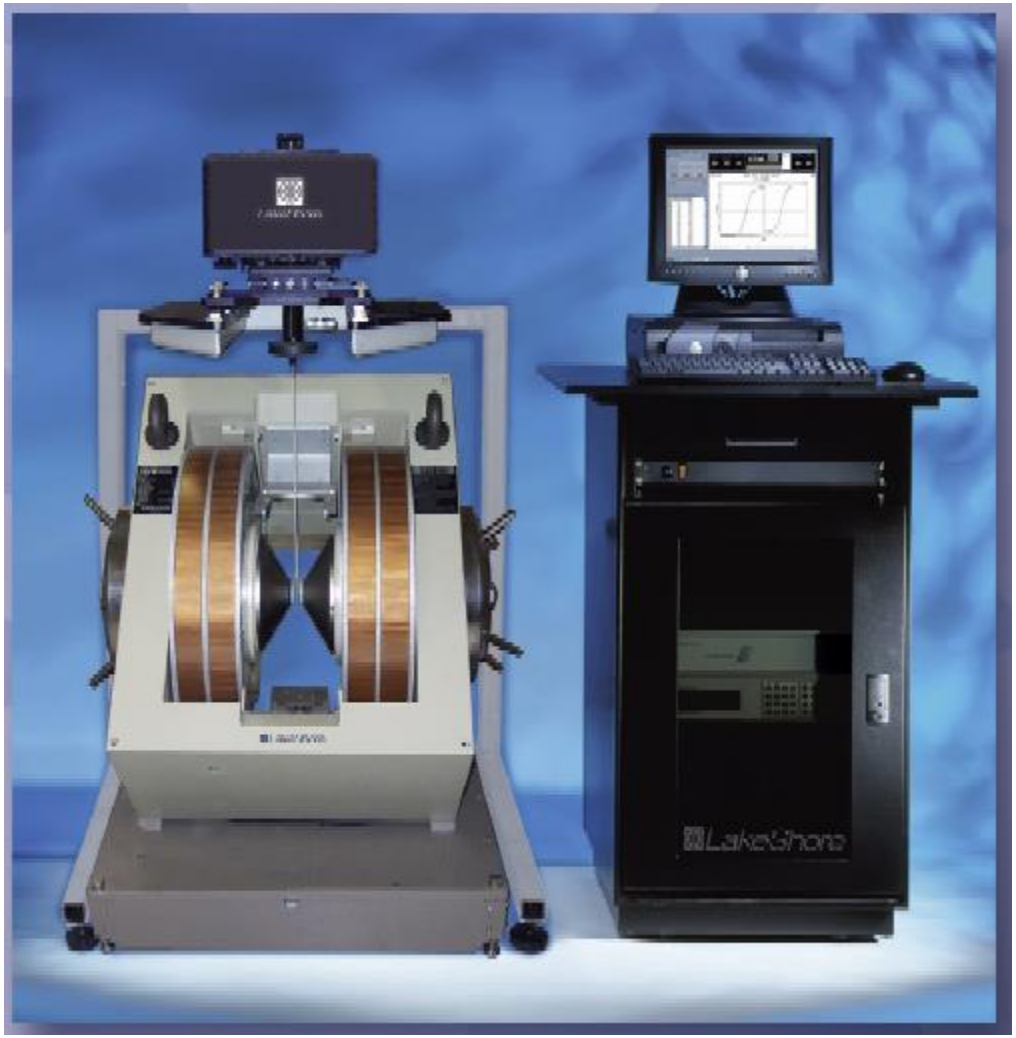
VSM – QD versalab



rms sensitivity: $<10^{-6}$ emu **or** 0.5 % with 1 sec. averaging

Relative Noise: Larger of (6×10^{-7} emu + 3×10^{-7} emu/tesla)/√Hz or 0.5%/√Hz

Lakeshore cryotronics VSM



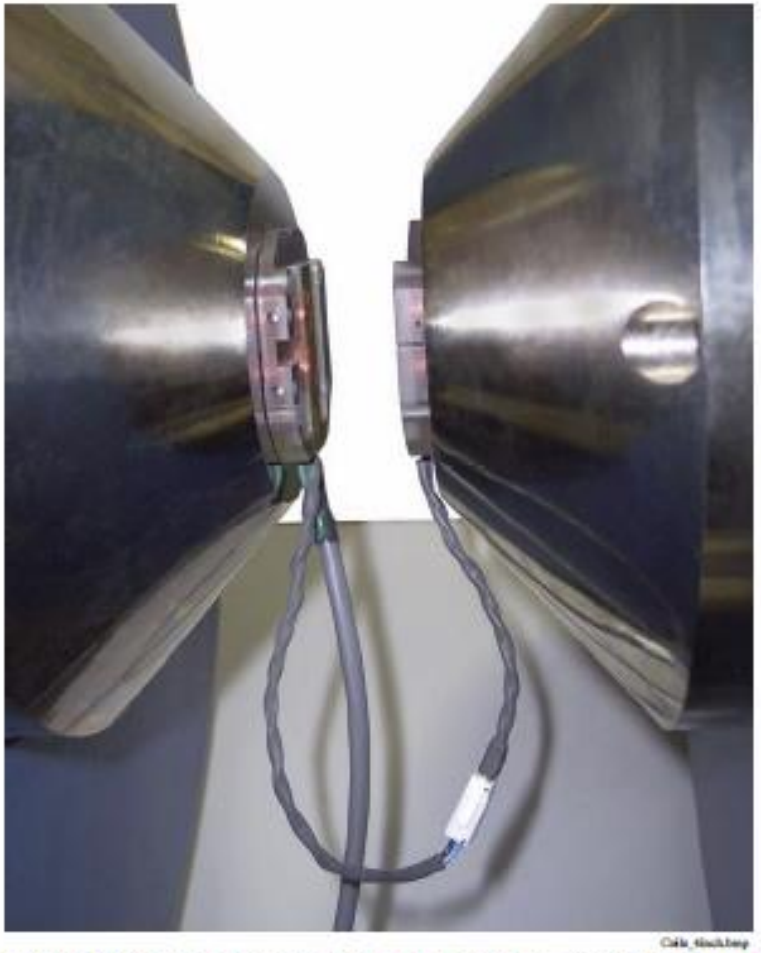
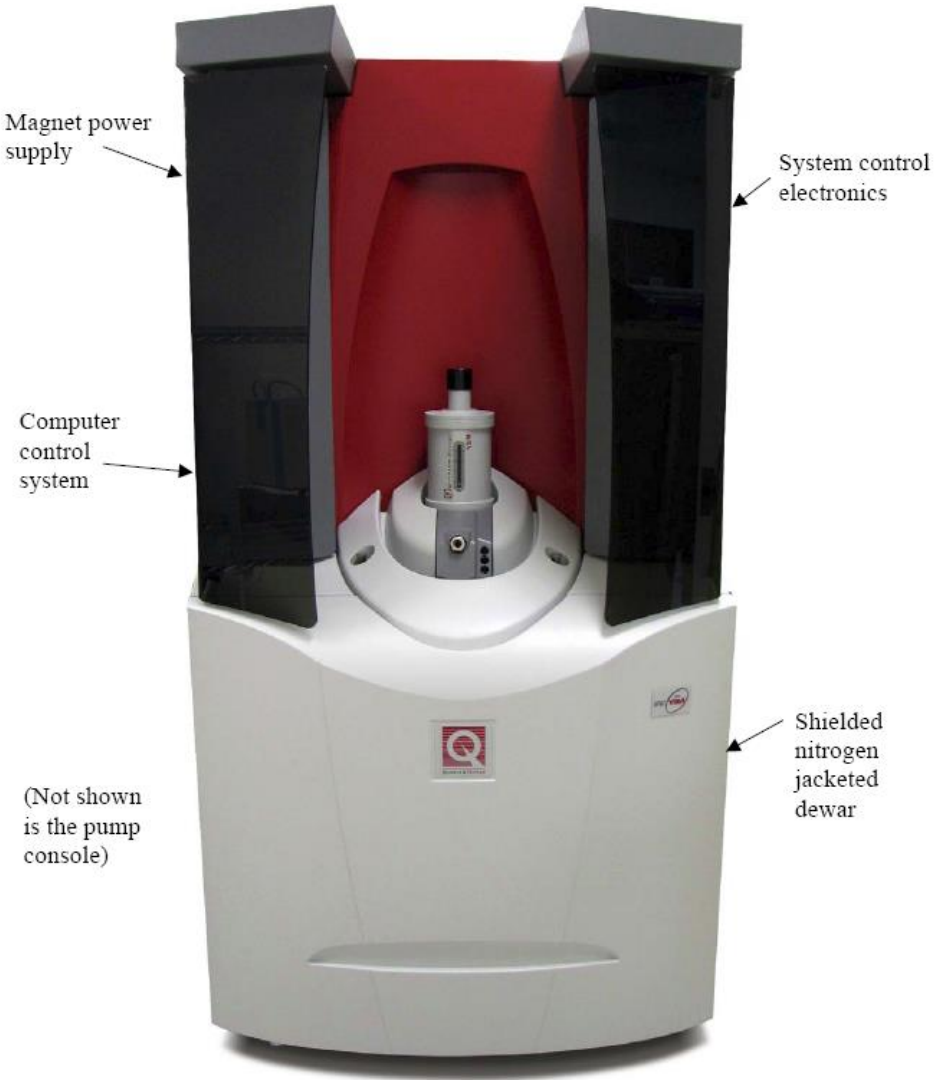


Figure 2-17. Typical 1-Inch Coils on 4-Inch Magnet Pole Cap with 2-Inch Face

QD SQUID-VSM magnetometer



Magnetization measurements

Feature: SQUID based VSM FastLab

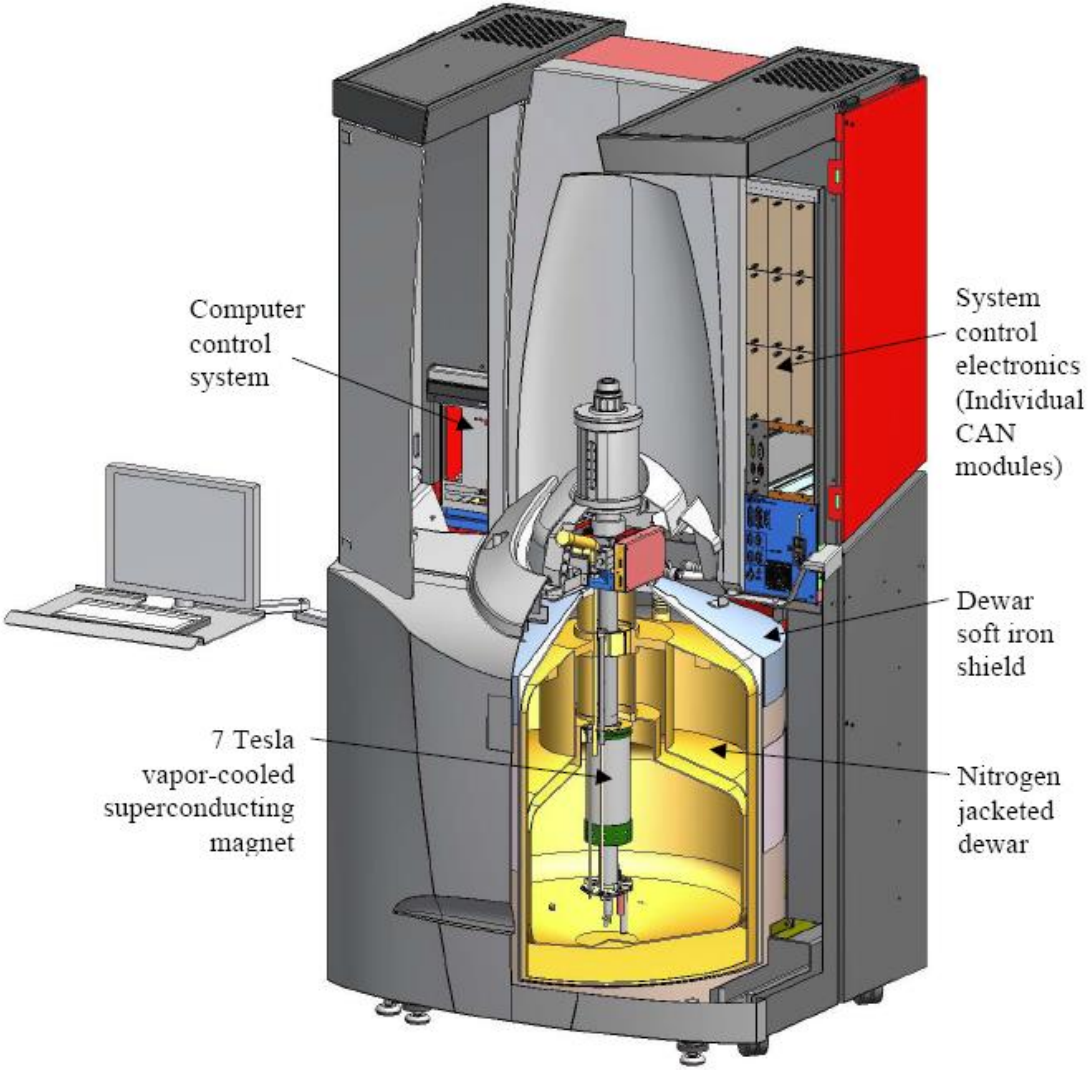
Maximum DC moment: 10 emu

Sensitivity: < 2,500 Oe < 1 x 10^{-8} emu (with less than 10 second averaging)

> 2,500 Oe < 8 x 10^{-8} emu (with less than 10 second averaging)

Variable drive amplitude: 0.1 to 5 mm (peak-to-peak)

Variable drive frequency: 5 to 80 Hz (Optimized for 28 Hz)



Cutaway view of the MPMS SQUID VSM.

QD MPMS

System Design

The MPMS family of magnetometers offers excellent sensitivity, field uniformity, repeatability, accuracy and reliability. Four models are available with magnetic field ranges from ±1 Tesla to ±7 Tesla, including ultra-low field options and AC/DC measurement capabilities. The modular MPMS design integrates a SQUID detection system, a precision temperature control unit residing in the bore of a high-field superconducting magnet, and a sophisticated computer operating system. Powerful software makes measurements, data collection and analysis quick and easy. Window-menu prompts and options lead the user through each step. When keyboard input is required, the system specifies what parameters should be entered, the appropriate units, and the allowable range of values.

